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Title 14—Aeronautics and Space
CHAPTER I—FEDERAL AVIATION ADMINISTRATION, DEPARTMENT OF TRANSPORTATION

[Docket No. 11011; Amdt. No. ~~22~~-14]

**PART 23—AIRWORTHINESS STANDARDS:
NORMAL, UTILITY, AND ACROBATIC
CATEGORY AIRPLANES**

Type Certification Standards

The purpose of these amendments is to update and improve the airworthiness standards applicable to airplanes type certificated under Part 23 of the Federal Aviation Regulations.

These amendments are based on a notice of proposed rulemaking published in the **FEDERAL REGISTER** (36 FR 8398) on May 5, 1971, and circulated as Notice 71-13, dated April 27, 1971.

A number of comments were received in response to Notice 71-13. Based on those comments and upon further review within the FAA, a number of changes have been made to the proposed rules. Those changes and the FAA's disposition of the relevant comments are discussed below. In addition, various non-substantive changes of a clarifying and editorial nature have been made. In general, comments received that were beyond the scope of the notice are not discussed but will be retained for consideration in connection with other rulemaking projects as appropriate.

(As published in the Federal Register
/38 F.R. 31816/ on November 19, 1973)

Interested persons have been afforded an opportunity to participate in the making of these amendments, and due consideration has been given to all matter presented. Except as modified by the following discussion, the reasons for these amendments are those set forth in Notice 71-13.

The notice proposed to amend § 23.143 (a) (5) to make it clear that the provisions of that section apply during landing with wing flaps extended as well as retracted. One commentator, while agreeing with the proposed amendment, suggested that the regulations should specify the speeds and techniques that must be used in showing compliance with the regulation. The FAA does not agree. The speeds and techniques necessary for the safe operation of the airplanes that must be furnished by the applicant under the provisions of § 23.1585 (a) are those that are used in showing compliance with § 23.143 (a) (5).

One commentator recommended that the words "or the minimum trim speed whichever is higher" be deleted from the proposed revision of § 23.145 (b) (1). The commentator stated that (1) a speed very much higher than $1.4 V_{S_0}$, even above the flap placard speed, could be required, and (2) the requirement that a transition be made from $1.4 V_{S_0}$ to $1.4 V_{S_0}$ is inaccurate and difficult to interpret if the minimum trim speed were used for the initial condition. The FAA does not agree with the commentator's first contention. Under § 23.161, the airplane must be shown to be trimmable in power-on descent conditions. An airplane which meets the trim conditions specified in § 23.161 will be trimmable at a speed not greatly in excess of $1.4 V_{S_0}$ and that minimum trim speed would never approach the flap placard speed. With respect to the commentator's second point, the proposal has been revised to provide for a transition from the minimum trim speed to a speed equal to V_{S_0} increased by the percentage that the minimum trim speed at the initial condition exceeded V_{S_0} . In addition, § 23.145 (b) (2) has been revised to more clearly state the requirement for the condition where the flaps are initially extended.

The notice proposed to add a new § 23.153 concerning control during landings. One comment stated that the proposed requirement should not apply to airplanes weighing 6,000 pounds or less maximum weight. The FAA agrees. The requirement, as stated in the notice, is necessitated by the advent of high performance aircraft that are being certificated under Part 23, and none of those aircraft are 6,000 pounds or less in maximum weight. The proposal has been revised accordingly. The commentator also suggested that the proposal be changed to reference only the forces prescribed for temporary application in § 23.143 (c). The FAA does not agree. The pilot's ability to maintain proper control during the maneuver may depend on the magnitude of prolonged forces encountered as well as temporary forces. Therefore, both the prolonged and temporary forces should be considered.

A comment was received contending that the use of extrapolation as provided in proposed § 23.155 (c) would not be appropriate in all cases. The FAA agrees. However, under § 23.21 (a) (1) extrapolations may only be used to show compliance if the result of those calculations would be equal in accuracy to the results of testing. Testing will be required if the stick force per g gradient indicates that extrapolation will not lead to an accurate result. Proposed § 23.155 (b) has been revised to more clearly indicate that two separate test conditions must be met, and that the turning flight condition is to be accomplished at the maximum level flight trim speed but not exceeding V_{W_2} or V_{M_0}/M_{M_0} , whichever is appropriate.

Proposed new § 23.157 (a) (3) has been revised to more clearly indicate that multiengine airplanes need only show compliance with the requirements of that subparagraph with one engine inoperative and the propeller in the minimum drag position, and the other engines at maximum continuous power or thrust.

One commentator suggested that proposed § 23.201 should be revised to provide an expanded definition of "stall" that would make that section consistent with § 25.201. The FAA does not agree. The airworthiness standards in Parts 23 and 25 are not the same and service experience with airplanes certificated under Part 23 indicates that for those airplanes there should be a different definition of "stall" than that which is contained in Part 25. Proposed § 23.201 (f) (5) has been deleted since the requirement is already contained in § 23.21 (a), and proposed § 23.201 (f) (7) is redesignated herein as § 23.201 (f) (6).

Two comments were received with respect to proposed § 23.203. The first recommended the inclusion of bank angle limits for turning flight stall demonstrations and the inclusion of a power-off condition for both turning flight stall and accelerated entry stall demonstrations. The inclusion of a power-off condition is beyond the scope of the notice. However, the comment is appreciated and it will be considered in connection with a future rule making action. Based on further consideration of the proposal in view of the comment received, the proposal has been revised to specify bank angle limits in place of the more general term "uncontrollable rolling tendencies" used in the proposal. The second comment requested that, for clarity, the title of the section be changed to "turning flight" and that the term "Normal Entry Stall" be used in place of "turning flight stall." The FAA does not agree. The title as proposed is more descriptive of the stated condition than the title suggested, and the term "turning flight stall" has been used previously in § 23.203 with no confusion. The commentator also suggested, and the FAA agrees, that for clarity the initial condition for showing compliance with the accelerated entry stall requirement should be specified and that that initial condition should be the same as that required for turning flight

stall demonstrations. The proposal has been revised accordingly. The requirement specified in proposed § 23.203 (b) (3) (vi) has been deleted since the requirements is contained in § 23.21 (a).

Proposed § 23.205 (b) (7) has also been deleted since the requirement is contained in § 23.21 (a). Proposed § 23.205 (b) (6) has been revised to include an alternate requirement for certain aircraft that are not capable of maintaining level flight with the critical engine inoperative. In addition, proposed § 23.205 (c) has been deleted since proposed §§ 23.205 (a) and 23.205 (b) adequately cover the matter proposed in paragraph (c).

The proposed amendment to § 23.307 has been withdrawn. It appears upon further review that proposed §§ 23.571 (a) and 23.572 (a) adequately cover the matter proposed in the proposed amendment to § 23.307.

One commentator requested that the prescribed flight conditions of proposed § 23.427 (a) be detailed as in present § 23.427. The FAA agrees and the section has been so revised. In addition, proposed § 23.427 (b) (2) has been redesignated § 23.427 (c) in order to clarify the requirement of that subsection. The commentator also suggested that yawing and slipstream effects should not be taken into consideration in complying with proposed § 23.427 (a). The FAA does not agree and no reason for the suggested change was presented by the commentator.

A comment questioned the reference in proposed § 23.445 (c) to "vertical surfaces in paragraph (b)." The proposed subsection is clear and the reference is correct.

In response to a comment received, §§ 23.507 (a) and 23.509 (c) (1) have been editorially revised for greater clarity. In addition, proposed § 23.507 (b) has been revised to state more clearly the requirement of that paragraph. One commentator objected to proposed § 23.507 (a) (2) (ii) as being too severe and suggested that 0.2 times the vertical static reactions be used for the fore, aft, and lateral load factors. The FAA does not agree. The commentator has not submitted, and the FAA is unaware of, any data to substantiate the claim of severity.

The notice explained that the purpose of the proposed amendment to § 23.572 (a) (1) was to make it clear that, for a fatigue evaluation, the use of analysis alone is acceptable only under certain specified circumstances. The word "only" has accordingly been inserted in that section to make the requirement clear. One commentator suggested that the section be revised to require fail safe structure in fatigue critical areas. The FAA does not agree. The comment is beyond the scope of the notice and a revision of the section, as suggested, would place an unnecessary restriction on aircraft design.

One commentator requested that proposed § 23.787 (c) be revised to require a total restraint capability of at least 9 g's for the structure and cargo restraint for the situation covered by the subsection.

The FAA does not agree. Service experience does not indicate that a 9g restraining capability is necessary, under the conditions specified in proposed § 23.787 (c), for all aircraft certificated under Part 23 irrespective of the type of operation for which the aircraft will be used.

In response to a comment received, proposed § 23.841(a) has been revised by deleting the word "reasonably" to be consistent with proposed § 23.1309. In addition, the second sentence of proposed § 23.841(b) (6) has been deleted since the sentence only stated an acceptable means of compliance and was not regulatory in nature.

A revision has been made to proposed § 23.853(d) to make it clear that the required protection is against any hazard rather than merely against damage.

The notice proposed to add a new § 23.865 concerned with fire protection of flight controls and other flight structure. One comment requested that flight structures be deleted from the provisions of the fire protection requirement on the asserted ground that experience does not indicate a need for such protection. The FAA does not agree. The commentator has submitted no data to substantiate its claim, and the information available to the FAA indicates, to the contrary, that flight structure in engine compartments must be protected against fire damage to ensure safe operation. However, the requirement has been clarified by specifying engine mounts as one type of flight structure which must meet the requirements of the section.

Proposed § 23.903(c) has been revised to make clear that the failure or malfunction of a fuel tank, if only one fuel tank is installed in the airplane, is not a condition making the provisions of that paragraph applicable. In addition, it should be noted that the language of § 23.903 as it existed prior to this amendment has been designated as § 23.903(a), and a paragraph (b) has been reserved.

The notice proposed to add a new § 23.929 concerned with icing protection of propellers and other components of complete engine installations. The title of the new section has been revised to be more descriptive of the purpose. One commentator suggested that instead of excluding wooden propellers from the requirement of the section, as proposed, fixed pitch propellers should be excluded. The FAA does not agree. The commentator did not present, and the FAA is unaware of, any data or service experience to indicate that fixed pitch propellers as a class are free from icing problems.

The Notice included a proposed amendment of § 23.987(d) to provide requirements for fuel tanks installed in personnel compartments of single engine airplanes without regard to the capacity of the tanks. A clarifying revision has been added to indicate that the required venting and draining must be to the exterior of the airplane. The Notice also included a proposed additional requirement intended to prevent fuel leakage under minor crash landing conditions.

Based on further review following a comment requesting that standards be established for rupture resistance of certain specific tanks, the requirement that tanks be rupture resistant is considered unnecessary and has been deleted since it is included in the requirement that the tanks retain fuel.

One commentator requested that the proposed new § 23.995(f) be clarified to require that check valves be constructed or other provisions made to preclude "inadvertent" incorrect assembly or connection of the valve. The FAA does not agree. The commentator's proposed clarification would permit the use of instructions on the valve to comply with the proposed requirement; however, service experience indicates that instructions on a valve are not adequate to preclude incorrect assembly or connection. The wording of the proposal has been revised to more clearly state the intent of the requirement.

The proposed new § 23.1093(c) has been withdrawn. It appears upon further consideration that the present rules adequately cover the matter proposed.

The Notice included proposals to revise present § 23.1141(e) to include reciprocating engines and to add a new § 23.1141(f) concerning fire resistance of powerplant controls. Based upon further consideration, the proposed revision of § 23.1141(e) has been withdrawn. The control system redundancy presently required for turbine engine powered aircraft is due to system complexities. However the control systems of reciprocating engines lack the complexity to necessitate such redundancy. Proposed § 23.1141(f) has been revised to require fire resistance only of those powerplant controls to be operated in the event of fire that are installed in the engine compartment. Those controls outside the engine compartment would not be subjected to immediate fire damage and need not be fire resistant.

The Notice included a proposed new § 23.1182 concerned with the construction and location of components, lines, and fittings behind engine firewalls. The proposal has been revised to use the terms "behind the firewall" in place of "safe side of the firewall" and "engine side" in place of "inner surface" for clarity and to be consistent with § 23.967. In addition, the proposal has been revised to explicitly exclude electrical equipment from the requirements of this section since electrical equipment must meet the requirements of § 23.1351(e). One commentator suggested that the proposal be revised to reduce from 15 minutes to 5 minutes the time the firewall is subjected to a flame temperature of not less than 2,000° F, since in proposed § 23.1351(e) 5 minutes is specified for electrical equipment. The FAA does not agree. Electrical equipment covered under proposed § 23.1351(e) has a different requirement than that proposed in § 23.1182 because fire damage to electrical equipment covered by proposed § 23.1351(e) is not as critical to the safe operation of the air-

plane as damage to the components, lines, and fittings covered by proposed § 23.1182.

Proposed § 23.1183(a) has been revised to require that components, lines, and fittings carrying flammable fluids, gas, or air in areas subject to engine fire conditions must be at least fire resistant. This indicates more clearly that fire proof materials may be used.

One commentator requested that proposed § 23.1189(c) be revised to permit ground check of valve settings in lieu of the indication to the flight crew that would be required by the proposal. The commentator states that: (1) Indicator lights would be required and that there may already be too many indicator lights in the cockpit, and (2) the valves presently in use have a record of safety. The FAA does not agree. A means of indication other than by indicator lights may be used, and since these valves may be operated from the cockpit there must be an indication to the crew of their position if the valves are ever to be used in extinguishing a fire.

The Notice included a proposed change to § 23.1301(a)(1) concerned with the performance of installed equipment. One commentator requested that the proposal be revised to refer to the most adverse likely condition instead of any foreseeable operating condition. The FAA agrees that as proposed an undue burden would be placed on the manufacturer. Accordingly, the proposal has been revised to refer to "probable" operating conditions. In response to another comment recommending that the proposal be changed to cover all required equipment, it has been revised to more clearly define the equipment subject to the requirement. With reference to a further suggestion that advisory material on the requirement be published, such advisory material has not been found necessary in the past and there is no present indication that it will be necessary under the amended rule.

As recommended in the comments, proposed §§ 23.1305 (s) and (t) have been redesignated §§ 23.1305 (q) and (r), respectively, in order to be consistent with the rest of § 23.1305. Proposed § 23.1305 (s), now § 23.1305 (q), has been revised to clarify that the only oil pressure warning means required is for low oil pressure.

The Notice proposed to add a new § 23.1309 concerned with the reliability of equipment, systems, and installations. Two comments were received with respect to the proposed section. One commentator suggested that advisory material would be desirable. While such action would not be regulatory in nature, the FAA will consider the preparation of advisory material in the future. The other commentator objected to the proposed section on the ground that it could be interpreted to require the complete elimination of radio interference which is impossible. That is not the intent and the FAA does not agree with that interpretation. Similarly worded requirements already exist in the Federal Aviation Regulations and have not been misinterpreted to place an undue burden on

manufacturers. However, based on further review, the proposed section has been revised to require that equipment installed in single engine airplanes be designed to minimize hazards to the airplane in the event of a probable malfunction or failure.

Proposed § 23.1321(d) (4) has been revised to provide that the magnetic direction indicator required by § 23.1303(c) need not be mounted in the instrument panel position proposed. The proposed requirement might create difficulties in calibrating and adjusting those indicators in the proposed position.

The proposed revisions to §§ 23.1323, 23.1501, and 23.1545, have been withdrawn pending a more comprehensive review by the FAA of the presently allowable airspeed indicating system error, and of the effect that the proposed changes to §§ 23.1323, 23.1501, and 23.1545 would have on that allowable error.

Proposed § 23.1351(e) has been revised by inserting the phrase "behind the firewall" in place of "outside a fire zone" and the phrase "in the engine compartment" in place of "in a designated fire zone" since airplanes certificated under Part 23 are not required to have designated fire zones.

Proposed § 23.1419(c) has been revised to permit, where applicable because of similarity of designs, the use of analysis and tests performed for the type certification of another type certificated aircraft for showing compliance with that section.

Proposed § 23.1435(a) (2) has been revised to indicate that the required indication must be to the flight crew.

Finally, proposed §§ 23.903 (d), (e) (1) and (3), 23.1419(a) and 23.1557(c) (2) have been revised to provide for the use of approved manual material to present the required information to the flight crew.

In consideration of the foregoing, Part 23 of the Federal Aviation Regulations is amended, effective December 20, 1973, as follows:

§ 23.143 [Amended]

1. Section 23.143(a) (5) is amended by inserting after the word "off", in the parenthetical statement, the words "with the wing flaps extended and retracted."

2. Section 23.145 is amended as follows:

a. By striking out the reference to "(4), or (5)" and inserting "or (4)" in place thereof in paragraphs (a) (2), (b) (4) and (6).

b. By amending paragraph (b) (1) and (2), and paragraph (d) to read as follows:

§ 23.145 Longitudinal control.

(b)

(1) With power off, flaps retracted, and the airplane trimmed at 1.4V_s or the minimum trim speed, whichever is higher, extend the flaps as rapidly as possible and allow the airspeed to transition from 1.4V_s to 1.4V_{s0}, or, if appropriate, from the minimum trim speed to a speed equal to V_s, increased by the

same percentage that the minimum trim speed at the initial condition was greater than V_s.

(2) With power off, flaps extended, and the airplane trimmed at 1.4V_s or the minimum trim speed, whichever is higher, retract the flaps as rapidly as possible and allow the airspeed to transition from 1.4V_s to 1.4V_{s0}, or, if appropriate, from the minimum trim speed to a speed equal to 1.4V_s, increased by the same percentage that the minimum trim speed at the initial condition was greater than V_s.

(d) It must be possible, with a pilot control force of not more than 10 pounds, to maintain a speed of not more than the speed determined in accordance with § 23.161(c) (4), during a power-off glide with landing gear and wing flaps extended.

3. A new § 23.153 is added to read as follows:

§ 23.153 Control during landings.

For an airplane that has a maximum weight of more than 6,000 pounds, it must be possible, while in the landing configuration, to safely complete a landing without encountering forces in excess of those prescribed in § 23.143(c) following an approach to land:

(a) At a speed 5 knots less than the speed used in complying with § 23.75 and with the airplane in trim or as nearly as possible in trim;

(b) With neither the trimming control being moved throughout the maneuver nor the power being increased during the landing flare; and

(c) With the thrust settings used in demonstrating compliance with § 23.75.

4. A new § 23.155 is added to read as follows:

§ 23.155 Elevator control force in maneuvers.

(a) The elevator control force needed to achieve the positive limit maneuvering load factor may not be less than:

(1) For wheel controls, W/100 (where W is the maximum weight) or 20 pounds, whichever is greater, except that it need not be greater than 50 pounds; or

(2) For stick controls, W/140 (where W is the maximum weight) or 15 pounds, whichever is greater, except that it need not be greater than 35 pounds.

(b) The requirement of paragraph (a) of this section must be met with wing flaps and landing gear retracted under each of the following conditions:

(1) At 75 percent of maximum continuous power for reciprocating engines, or the maximum power or thrust selected by the applicant as an operating limitation for use during cruise for reciprocating or turbine engines.

(2) In a turn, after the airplane is trimmed with wings level at the minimum speed at which the required normal acceleration can be achieved without stalling, and at the maximum level flight trim speed except that the speed may not

exceed V_N or V_{MO}/M₀ whichever is appropriate.

(c) Compliance with the requirements of this section may be demonstrated by measuring the normal acceleration that is achieved with the limiting stick force or by establishing the stick force per gradient and extrapolating to the appropriate limit.

5. A new § 23.157 is added to read as follows:

§ 23.157 Rate of roll.

(a) *Takeoff.* It must be possible, using a favorable combination of controls, to roll the airplane from a steady 30-degree banked turn through an angle of 60 degrees, so as to reverse the direction of the turn within:

(1) For an airplane of 6,000 pounds or less maximum weight, 5 seconds from initiation of roll; and

(2) For an airplane of over 6,000 pounds maximum weight,

$$\frac{W+500}{1,300}$$

seconds, where W is the weight in pounds.

(b) The requirement of paragraph (a) must be met when rolling the airplane in either direction in the following condition:

(1) Flaps in the takeoff position;

(2) Landing gear retracted;

(3) For a single engine airplane, at maximum takeoff power or thrust; and for a multiengine airplane, with the critical engine inoperative, the propeller in the minimum drag position, and the other engines at maximum continuous power or thrust; and

(4) The airplane trimmed at 1.2V_s, or as nearly as possible in trim for straight flight.

(c) *Approach.* It must be possible, using a favorable combination of controls, to roll the airplane from a steady 30-degree banked turn through an angle of 60 degrees, so as to reverse the direction of the turn within:

(1) For an airplane of 6,000 pounds or less maximum weight, 4 seconds from initiation of roll; and

(2) For an airplane of over 6,000 pounds maximum weight,

$$\frac{W+2,800}{2,200}$$

seconds, where W is the weight in pounds.

(d) The requirement of paragraph (c) must be met when rolling the airplane in either direction in the following conditions:

(1) Flaps extended;

(2) Landing gear extended;

(3) All engines operating at idle power or thrust and with all engines operating at the power or thrust for level flight; and

(4) The airplane trimmed at the speed that is used in determining compliance with § 23.75.

6. Section 23.161 is amended by deleting paragraph (c) (5) and by amending paragraphs (c) (3) and (4) to read as follows:

§ 23.161 Trim.

(c) Longitudinal trim.

(3) A power approach with a 3 degree angle of descent, the landing gear extended, flaps retracted, and:

(i) For an airplane of 6,000 pounds or less maximum weight, a speed between 1.3Vs, and 1.5Vs; or

(ii) For an airplane of more than 6,000 pounds maximum weight, a speed of 1.4Vs; and

(4) A power approach with a 3 degree angle of descent, the landing gear extended, and:

(i) For an airplane of 6,000 pounds or less maximum weight, a speed between 1.3Vs, and 1.5Vs, with flaps extended; or

(ii) For an airplane of more than 6,000 pounds maximum weight, the speed and flap position used in showing compliance with § 23.75(a).

(5) (Reserved.)

7. Paragraph (a) of § 23.173 is amended to read as follows:

§ 23.173 Static longitudinal stability.

(a) A pull must be required to obtain and maintain speeds below the specified trim speed and a push required to obtain and maintain speeds above the specified trim speed. This must be shown at any speed that can be obtained, except that speeds requiring a control force in excess of 40 pounds or speeds above the maximum allowable speed or below the minimum speed for steady unstalled flight, need not be considered.

§ 23.175 [Amended]

8. Section 23.175 is amended as follows:

a. By striking out the parenthetical statement in paragraph (b) (2).

b. By striking out of the parenthetical statement in paragraph (c) the words "nor speeds that require a stick force of more than 40 pounds."

c. By amending paragraph (d) by striking out the words "and the stick force may not exceed 40 pounds."

d. By amending paragraph (d) (3) by striking out the words "and (5)."

e. By amending paragraph (d) (4) to read: "Both power off and enough power to maintain a 3 degree angle of descent."

9. Section 23.201 is amended to read as follows:

§ 23.201 Wings level stall.

(a) For an airplane with independently controlled roll and directional controls, it must be possible to produce and to correct roll by unreversed use of the rolling control and to produce and to correct yaw by unreversed use of the directional control, up to the time the airplane pitches.

(b) For an airplane with interconnected lateral and directional controls (2 controls) and for an airplane with only one of these controls, it must be possible to produce and correct roll by unreversed use of the rolling control without pro-

ducing excessive yaw, up to the time the airplane pitches.

(c) The wing level stall characteristics of the airplane must be demonstrated in flight as follows: The airplane speed must be reduced with the elevator control until the speed is slightly above the stalling speed, then the elevator control must be pulled back so that the rate of speed reduction will not exceed one knot per second until a stall is produced, as shown by an uncontrollable downward pitching motion of the airplane, or until the control reaches the stop. Normal use of the elevator control for recovery is allowed after the pitching motion has unmistakably developed.

(d) Except where made inapplicable by the special features of a particular type of airplane, the following apply to the measurement of loss of altitude during a stall:

(1) The loss of altitude encountered in the stall (power on or power off) is the change in altitude (as observed on the sensitive altimeter testing installation) between the altitude at which the airplane pitches and the altitude at which horizontal flight is regained.

(2) If power or thrust is required during stall recovery the power or thrust used must be that which would be used under the normal operating procedures selected by the applicant for this maneuver. However, the power used to regain level flight may not be applied until flying control is regained.

(e) During the recovery part of the maneuver, it must be possible to prevent more than 15 degrees of roll or yaw by the normal use of controls.

(f) Compliance with the requirements of this section must be shown under the following conditions:

(1) Wing flaps: Full up, full down, and intermediate, if appropriate.

(2) Landing Gear: Retracted and extended.

(3) Cowl Flaps: Appropriate to configuration.

(4) Power: Power or thrust off, and 75 percent maximum continuous power or thrust.

(5) Trim: 1.5Vs, or at the minimum trim speed, whichever is higher.

(6) Propeller: Full increase rpm position for the power off condition.

10. Section 23.203 is amended to read as follows:

§ 23.203 Turning flight and accelerated stalls.

Turning flight and accelerated stalls must be demonstrated in flight tests as follows:

(a) Established and maintain a coordinated turn in a 30 degree bank. Reduce speed by steadily and progressively tightening the turn with the elevator until the airplane is stalled or until the elevator has reached its stop. The rate of speed reduction must be constant, and:

(1) For a turning flight stall, may not exceed one knot per second; and

(2) For an accelerated stall, be 3 to 5 knots per second with steadily increasing normal acceleration.

(b) When the stall has fully developed or the elevator has reached its stop, it must be possible to regain level flight without:

(1) Excessive loss of altitude;

(2) Undue pitchup;

(3) Uncontrollable tendency to spin;

(4) Exceeding 60 degree of roll in either direction from the established 30 degree bank; and

(5) For accelerated entry stalls, without exceeding the maximum permissible speed or the allowable limit load factor.

(c) Compliance with the requirements of this section must be shown with:

(1) Wing flaps: Retracted and fully extended for turning flight and accelerated entry stalls, and intermediate, if appropriate, for accelerated entry stalls;

(2) Landing gear: Retracted and extended;

(3) Cowl flaps: Appropriate to configuration;

(4) Power: 75 percent maximum continuous power; and

(5) Trim: 1.5Vs, or minimum trim speed, whichever is higher.

11. Section 23.205 is amended to read as follows:

§ 23.205 Critical engine inoperative stalls.

(a) A multiengine airplane may not display any undue spinning tendency and must be safely recoverable without applying power to the inoperative engine when stalled. The operating engines may be throttled back during the recovery from stall.

(b) Compliance with paragraph (a) of the section must be shown with:

(1) Wing flaps: Retracted.

(2) Landing gear: Retracted.

(3) Cowl flaps: Appropriate to level flight critical engine inoperative.

(4) Power: Critical engine inoperative and the remaining engine(s) at 75 percent maximum continuous power or thrust or the power or thrust at which the use of maximum control travel just holds the wings laterally level in the approach to stall, whichever is lesser.

(5) Propeller: Normal inoperative position for the inoperative engine.

(6) Trim: Level flight, critical engine inoperative, except that for an airplane of 6,000 pounds or less maximum weight that has a stalling speed of 61 knots or less and cannot maintain level flight with the critical engine inoperative, the airplane must be trimmed for straight flight, critical engine inoperative, at a speed not greater than 1.5Vs.

12. Section 23.427 is amended to read as follows:

§ 23.427 Unsymmetrical loads.

(a) Horizontal tail surfaces and their supporting structure must be designed for unsymmetrical loads arising from yawing and slipstream effects, in combination with the loads prescribed for the flight conditions set forth in §§ 23.421 through 23.425.

(b) In the absence of more rational data for airplanes that are conventional

in regard to location of engines, wings, tail surfaces, and fuselage shape:

(1) 100 percent of the maximum loading from the symmetrical flight conditions may be assumed on the surface on one side of the plane of symmetry; and

(2) The following percentage of that loading must be applied to the opposite side:

Percent = $100 - 10(n - 1)$, where n is the specified positive maneuvering load factor, but this value may not be more than 80 percent.

(c) For airplanes that are not conventional (such as airplanes with horizontal tail surfaces having appreciable dihedral or supported by the vertical tail surfaces) the surfaces and supporting structures must be designed for combined vertical and horizontal surface loads resulting from each prescribed flight condition taken separately.

13. Paragraph (a) of § 23.441 is amended to read as follows:

§ 23.441 Maneuvering loads.

(a) As speeds up to V_A , the vertical tail surfaces must be designed to withstand the following conditions. In computing the tail loads, the yawing velocity may be assumed to be zero:

(1) With the airplane in unaccelerated flight at zero yaw, it is assumed that the rudder control is suddenly displaced to the maximum deflection, as limited by the control stops or by limit pilot forces.

(2) With the rudder deflected as specified in paragraph (a) (1) of this section, it is assumed that the airplane yaws to the resulting sideslip angle. In lieu of a rational analysis, an overswing angle equal to 1.3 times the static sideslip angle of paragraph (a) (3) of this section may be assumed.

(3) A yaw angle of 15 degrees with the rudder control maintained in the neutral position (except as limited by pilot strength).

14. Section 23.445 is amended by adding a new paragraph (c) to read as follows:

§ 23.445 Outboard fins.

(c) The end plate effects of outboard fins must be taken into account in applying the yawing conditions of § 23.441 and § 23.443 to the vertical surfaces in paragraph (b) of this section.

15. A new § 23.507 is added to read as follows:

§ 23.507 Jacking loads.

(a) The airplane must be designed for the loads developed when the aircraft is supported on jacks at the design maximum weight assuming the following load factors for landing gear jacking points at a three-point attitude and for primary flight structure jacking points in the level attitude:

(1) Vertical-load factor of 1.35 times the static reactions.

(2) Fore, aft, and lateral load factors of 0.4 times the vertical static reactions.

(b) The horizontal loads at the jack points must be reacted by inertia forces so as to result in no change in the direction of the resultant loads at the jack points.

(c) The horizontal loads must be considered in all combinations with the vertical load.

16. A new § 23.509 is added to read as follows:

§ 23.509 Towing loads.

The towing loads of this section must be applied to the design of tow fittings and their immediate attaching structure.

(a) The towing loads specified in paragraph (d) of this section must be considered separately. These loads must be applied at the towing fittings and must act parallel to the ground. In addition:

(1) A vertical load factor equal to 1.0 must be considered acting at the center of gravity; and

(2) The shock struts and tires must be in their static positions.

(b) For towing points not on the landing gear but near the plane of symmetry of the airplane, the drag and side tow load components specified for the auxil-

iary gear apply. For towing points located outboard of the main gear, the drag and side tow load components specified for the main gear apply. Where the specified angle of swivel cannot be reached, the maximum obtainable angle must be used.

(c) The towing loads specified in paragraph (d) of this section must be reacted as follows:

(1) The side component of the towing load at the main gear must be reacted by a side force at the static ground line of the wheel to which the load is applied.

(2) The towing loads at the auxiliary gear and the drag components of the towing loads at the main gear must be reacted as follows:

(i) A reaction with a maximum value equal to the vertical reaction must be applied at the axle of the wheel to which the load is applied. Enough airplane inertia to achieve equilibrium must be applied.

(ii) The loads must be reacted by airplane inertia.

(d) The prescribed towing loads are as follows, where W is the design maximum weight:

Tow point	Position	Load		
		Magnitude	No.	Direction
Main gear		0.25W per main gear unit.	1 2 3 4	Forward, parallel to drag axis. Forward, at 30° to drag axis. Aft, parallel to drag axis. Aft, at 30° to drag axis.
Auxiliary Gear	Swiveled forward	0.3W	5 6	Forward. Aft.
	Swiveled Aft		7 8	Forward. Aft.
	Swiveled 45° from forward	0.15W	9 10	Forward, in plane of wheel. Aft, in plane of wheel.
	Swiveled 45° from aft		11 12	Forward, in plane of wheel. Aft, in plane of wheel.

17. Paragraph (a) of § 23.571 is amended to read as follows:

§ 23.571 Pressurized cabin.

(a) A fatigue strength investigation, in which the structure is shown by analysis, tests, or both to be able to withstand the repeated loads of variable magnitude expected in service. Analysis alone is considered acceptable only when it is conservative and applied to simple structures.

18. Paragraph (a) (1) of § 23.572 is amended to read as follows:

§ 23.572 Wing and associated structure.

(a) . . .

(1) A fatigue strength investigation, in which the structure is shown by analysis, tests, or both, to be able to withstand the repeated loads of variable magnitude expected in service. Analysis alone is acceptable only when it is conservative and applied to simple structures.

19. Paragraph (b) of § 23.701 is amended to read as follows:

§ 23.701 Flap interconnection.

(b) If an interconnection is used in multiengine airplanes, it must be designed to account for the unsymmetrical loads resulting from flight with the engines on one side of the plane of symmetry inoperative and the remaining engines at takeoff power. For single-engine airplanes, and multiengine airplanes with no slipstream effects on the flaps, it may be assumed that 100 percent of the critical air load acts on one side and 70 percent on the other.

20. Section 23.771 is amended by striking out the word "and" at the end of paragraph (a) and by redesignating paragraph (b) as paragraph (c) and by adding a new paragraph (b) to read as follows:

§ 23.771 Pilot compartment.

(b) Where the flight crew are separated from the passengers by a partition, an opening or openable window or door must be provided to facilitate communication between flight crew and the passengers; and

21. Section 23.773 is amended as follows:

a. By striking out the word "and" at the end of paragraph (a) (1);

b. By deleting the period at the end of paragraph (a) (2), and inserting "; and" in place thereof; and

c. By adding a new paragraph (a) (3) to read as follows:

§ 23.773 Pilot compartment view.

(a) . . .

(3) Internal fogging of the windows covered under paragraph (a) (1) of this section can be easily cleared by each pilot unless means are provided to prevent fogging.

22. Section 23.787 is amended by amending paragraphs (b) and (c) and by adding new paragraphs (d) and (e) to read as follows:

§ 23.787 Cargo compartments.

(b) There must be means to prevent the contents of any cargo compartment from becoming a hazard by shifting, and to protect any controls, wiring, lines, equipment or accessories whose damage or failure would affect safe operations.

(c) Where the cargo compartment is located aft of occupants and separated from them by structure, there must be means within the cargo compartment to protect the occupants from injury by the contents of the cargo compartment when the ultimate forward inertia force is 4.5g.

(d) Cargo compartments must be constructed of materials which are at least flame resistant.

(e) Designs which provide for cargo to be carried in the same compartment with the occupants must have means to protect the occupants from injury under the ultimate inertia forces specified in § 23.561(b) (2).

23. Section 23.841 is amended to read as follows:

§ 23.841 Pressurized cabins.

(a) If certification for operation over 31,000 feet is requested, the airplane must be able to maintain a cabin pressure altitude of not more than 15,000 feet in event of any probable failure or malfunction in the pressurization system.

(b) Pressurized cabins must have at least the following valves, controls, and indicators, for controlling cabin pressure:

(1) Two pressure relief valves (at least one of which is the normal regulating valve) to automatically limit the positive pressure differential to a predetermined value at the maximum rate of flow delivered by the pressure source. The combined capacity of the relief valves must be large enough so that the failure of any one valve would not cause an appreciable rise in the pressure differential. The pressure differential is positive when the internal pressure is greater than the external.

(2) Two reverse pressure differential relief valves (or their equivalent) to automatically prevent a negative pressure

differential that would damage the structure. However, one valve is enough if it is of a design that reasonably precludes its malfunctioning.

(3) A means by which the pressure differential can be rapidly equalized.

(4) An automatic or manual regulator for controlling the intake or exhaust airflow, or both, for maintaining the required internal pressures and airflow rates.

(5) Instruments to indicate to the pilot the pressure differential, the absolute pressure in the cabin, and the rate of change of the absolute pressure.

(6) Warning indicators at the pilot station to indicate when the safe or preset pressure differential and absolute cabin pressure limits are exceeded.

(7) A warning placard for the pilot if the structure is not designed for pressure differentials up to the maximum relief valve setting in combination with landing loads.

(8) A means to stop rotation of the compressor or to divert airflow from the cabin if continued rotation of an engine-driven cabin compressor or continued flow of any compressor bleed air will create a hazard if a malfunction occurs.

24. Section 23.853 is amended to read as follows:

§ 23.853 Compartment interiors.

For each compartment to be used by the crew or passengers:

(a) The materials must be at least flame-resistant;

(b) If smoking is to be allowed:

(1) There must be an adequate number of self-contained ash trays;

(2) Where the crew compartment is separated from the passenger compartment, there must be an illuminated no smoking sign (or signs) controllable from a flight crew station and readable from each passenger seat to indicate when smoking is prohibited;

(c) If smoking is to be prohibited, there must be a placard so stating; and

(d) Lines, tanks, or equipment containing fuel, oil, or other flammable fluids may not be installed in such compartments unless adequately shielded, isolated, or otherwise protected so that any breakage or failure of such an item would not create a hazard.

25. A new § 23.865 is added to read as follows:

§ 23.865 Fire protection of flight controls and other flight structure.

Flight controls, engine mounts, and other flight structure located in the engine compartment must be constructed of fireproof material or shielded so that they will withstand the effect of a fire.

26. Section 23.903 is amended by designating the former requirement of § 23.903 as paragraph (a), reserving paragraph (b), and adding new paragraphs (c), (d), and (e) to read as follows:

§ 23.903 Engines.

(a) Each engine must be type certificated under Part 33.

(b) [Reserved]

(c) The powerplants must be arranged and isolated from each other to allow operation, in at least one configuration, so that the failure or malfunction of any engine, or the failure or malfunction (including destruction by fire in the engine compartment) of any system that can affect an engine (other than a fuel tank if only one fuel tank is installed), will not:

(1) Prevent the continued safe operation of the remaining engines; or

(2) Require immediate action by any crewmember for continued safe operation of the remaining engines.

(d) *Starting and stopping (piston engine).* The design of the installation must be such that risk of fire or mechanical damage to the engine or airplane, as a result of starting the engine in any conditions in which starting is to be permitted, is reduced to a minimum. Any techniques and associated limitations for engine starting must be established and included in the Airplane Flight Manual, approved manual material, or applicable operating placards. For multiengine airplanes, means must be provided for stopping and restarting each engine in flight. For single-engine airplanes, means must be provided for stopping the engine in flight after engine failure if overspeeding might be caused by windmilling of the propeller.

(e) *Starting and stopping (turbine engine).* Turbine engine installations must comply with the following:

(1) The design of the installation must be such that risk of fire or mechanical damage to the engine or the airplane, as a result of starting the engine in any conditions in which starting is to be permitted, is reduced to a minimum. Any techniques and associated limitations must be established and included in the Airplane Flight Manual, approved manual material, or applicable operating placards.

(2) Means must be provided for stopping combustion and rotation of any engine. All those components provided for compliance with this requirement, which are within any engine compartment, on the engine side of the firewall, must be at least fire resistant.

(3) It must be possible to restart an engine in flight. Any techniques and associated limitations must be established and included in the Airplane Flight Manual, approved manual material, or applicable operating placards.

(4) It must be demonstrated in flight that when restarting engines following a false start, all fuel or vapor is discharged in such a way that it does not constitute a fire hazard.

27. A new § 23.929 is added to read as follows:

§ 23.929 Engine installation ice protection.

Propellers (except wooden propellers) and other components of complete engine installations must be protected against the accumulation of ice as necessary to enable satisfactory functioning without appreciable loss of power when

operated in the icing conditions for which certification is requested.

28. Section 23.939 is amended by amending the heading and paragraph (b) to read as follows:

§ 23.939 Powerplant operating characteristics.

(b) No hazardous malfunction of the powerplant may occur when the airplane is operated at the negative acceleration within the flight envelope prescribed in § 23.333 that is most critical. This must be shown for the greatest duration expected for that acceleration.

29. Paragraph (d) of § 23.967 is amended and a new paragraph (e) is added to § 23.967 to read as follows:

§ 23.967 Fuel tank installation.

(d) No fuel tank may be installed in the personnel compartment of a multi-engine airplane. If a fuel tank is installed in the personnel compartment of a single-engine airplane, it must be isolated by fume and fuel-proof enclosures that are drained and vented to the exterior of the airplane. A bladder type fuel cell, if used, must have a retaining shell at least equivalent to a metal fuel tank in structural integrity.

(e) Fuel tanks must be designed, located, and installed so as to retain fuel:

(1) Under the inertia forces prescribed for the emergency landing conditions in § 23.561; and

(2) Under conditions likely to occur when an airplane lands either with its landing gear retracted or one landing gear collapsed, or when an engine mounting tears away.

30. A new § 23.979 is added to read as follows:

§ 23.979 Pressure fueling systems.

For pressure fueling systems, the following apply:

(a) Each pressure fueling system fuel manifold connection must have means to prevent the escape of hazardous quantities of fuel from the system if the fuel entry valve fails.

(b) An automatic shutoff means must be provided to prevent the quantity of fuel in each tank from exceeding the maximum quantity approved for that tank. This means must allow checking for proper shutoff operation before each fueling of the tank.

(c) A means must be provided to prevent damage to the fuel system in the event of failure of the automatic shutoff means prescribed in paragraph (b) of this section.

(d) All parts of the fuel system up to the tank which are subjected to fueling pressures must have a proof pressure of 1.33 times, and an ultimate pressure of at least 2.0 times, the surge pressure likely to occur during fueling.

31. A new paragraph (f) is added to § 23.995 to read as follows:

§ 23.995 Fuel valves and controls.

(f) Each check valve must be constructed, or otherwise incorporate provisions, to preclude incorrect assembly or connection of the valve.

32. Section 23.1017 is amended by amending paragraph (b) (1) and by adding a new paragraph (b) (5) to read as follows:

§ 23.1017 Oil lines and fittings.

(b) *Breather lines.*

(1) Condensed water vapor or oil that might freeze and obstruct the line cannot accumulate at any point.

(5) The breather outlet is protected against blockage by ice or foreign matter.

33. A new paragraph (d) is added to § 23.1027 to read as follows:

§ 23.1027 Propeller feathering system.

(d) Provision must be made to prevent sludge or other foreign matter from affecting the safe operation of the propeller feathering system.

34. A new paragraph (f) is added to § 23.1141 to read as follows:

§ 23.1141 Powerplant controls: General.

(f) The portion of each powerplant control located in the engine compartment that is required to be operated in the event of fire must be at least fire resistant.

35. A new paragraph (c) is added to § 23.1163 to read as follows:

§ 23.1163 Powerplant accessories.

(c) Each generator rated at or more than 6 kilowatts must be designed and installed to minimize the probability of a fire hazard in the event it malfunctions.

36. Under the heading Powerplant Fire Protection, a new § 23.1182 is added to read as follows:

§ 23.1182 Nacelle areas behind firewalls.

Components, lines, and fittings, except those subject to the provisions of § 23.1351(e), located behind the engine-compartment firewall must be constructed of such materials and located at such distances from the firewall that they will not suffer damage sufficient to endanger the airplane if a portion of the engine side of the firewall is subjected to a flame temperature of not less than 2000° F for 15 minutes.

37. Paragraph (a) of § 23.1183 is amended to read as follows:

§ 23.1183 Lines and fittings and components.

(a) Except as a provided in paragraph (b) of this section, each component, line, and fitting carrying flammable fluids, gas, or air in any area subject to engine fire conditions must be at least fire

resistant. Flexible hose assemblies (hose and end fittings) must be approved.

38. A new paragraph (c) is added to § 23.1189 to read as follows:

§ 23.1189 Shutoff means.

(c) Power operated valves must have means to indicate to the flight crew when the valve has reached the selected position and must be designed so that the valve will not move from the selected position under vibration conditions likely to exist at the valve location.

39. A new § 23.1192 is added to read as follows:

§ 23.1192 Engine accessory compartment diaphragm.

For aircooled radial engines, the engine power section and all portions of the exhaust system must be isolated from the engine accessory compartment by a diaphragm that meets the firewall requirements of § 23.1191.

40. Paragraph (a) (1) of § 23.1301 is amended to read as follows:

§ 23.1301 Function and installation.

(a) Each item of installed equipment required by this chapter or otherwise essential to safe operation must:

(1) Adequately perform its intended function under all probable operating conditions.

41. New paragraphs (q) and (r) are added to § 23.1305 to read as follows:

§ 23.1305 Powerplant instruments.

(q) A low oil pressure warning means for each turbine engine.

(r) An induction system air temperature indicator for each engine equipped with a preheater and having induction air temperature limitations which can be exceeded with preheat.

42. Under the heading "General", a new § 23.1309 is added to read as follows:

§ 23.1309 Equipment systems and installations.

(a) Each item of equipment, when performing its intended function, may not adversely affect:

(1) The response, operation, or accuracy of any equipment essential to safe operation; or

(2) The response, operation, or accuracy of any other equipment unless there is a means to inform the pilot of the effect.

(b) The equipment, systems, and installations of a multiengine airplane must be designed to prevent hazards to the airplane in the event of a probable malfunction or failure.

(c) The equipment, systems, and installations of a single-engine airplane must be designed to minimize hazards to the airplane in the event of a probable malfunction or failure.

43. Paragraph (a) of § 23.1321 is amended and a new paragraph (d) is added to read as follows:

§ 23.1321 Arrangement and visibility.

(a) Each flight, navigation, and power-plant instrument for use by any pilot must be plainly visible to him from his station with the minimum practicable deviation from his normal position and line of vision when he is looking forward along the flight path.

(d) For each airplane of more than 6,000 pounds maximum weight, the flight instruments required by § 23.1303, and as applicable, by Part 91 of this chapter must be grouped on the instrument panel and centered as nearly as practicable about the vertical plane of the pilot's forward vision. In addition:

(1) The instrument that most effectively indicates the attitude must be on the panel in the top center position;

(2) The instrument that most effectively indicates airspeed must be adjacent to and directly to the left of the instrument in the top center position;

(3) The instrument that most effectively indicates altitude must be adjacent to and directly to the right of the instrument in the top center position; and

(4) The instrument that most effectively indicates direction of flight, other than the magnetic direction indicator required by § 23.1303(c), must be adjacent to and directly below the instrument in the top center position.

44. Section 23.1351 is amended as follows:

a. By striking out the word "and" at the end of paragraph (b) (1) (i).

b. By striking out the period and adding the words "; and" at the end of paragraph (b) (1) (ii).

c. By striking out the word "and" at the end of paragraph (c) (2).

d. By striking out the period and adding the words "; and" at the end of paragraph (c) (3).

e. By adding a new paragraph (b) (1) (iii), a new paragraph (c) (4), and a new paragraph (e) to read as follows:

§ 23.1351 General.

(b) *Functions.* (1) . . .

(iii) So designed that the risk of electrical shock to crew, passengers, and ground personnel is reduced to a minimum.

(c) *Generating system.* . . .

(4) There must be a means to give immediate warning to the flight crew of

a failure of any generator.

(e) *Fire resistance.* Electrical equipment must be so designed and installed that in the event of a fire in the engine compartment, during which the surface of the firewall adjacent to the fire is heated to 2,000° F for 5 minutes or to a lesser temperature substantiated by the applicant, the equipment essential to continued safe operation, and located behind the firewall will function satisfactorily and will not create an additional fire hazard.

45. Section 23.1365 is amended by amending the heading and paragraph (b) to read as follows:

§ 23.1365 Electric cables and equipment.

(b) Each cable and associated equipment that would overheat in the event of circuit overload or fault must be at least flame resistant and may not emit dangerous quantities of toxic fumes.

46. Section 23.1419 is amended to read as follows:

§ 23.1419 Ice protection.

If certification with ice protection provisions is desired, compliance with the following requirements must be shown:

(a) The recommended procedures for the use of the ice protection equipment must be set forth in the Airplane Flight Manual or in approved manual material.

(b) An analysis must be performed to establish, on the basis of the airplane's operational needs, the adequacy of the ice protection system for the various components of the airplane. In addition, tests of the ice protection system must be conducted to demonstrate that the airplane is capable of operating safely in continuous maximum and intermittent maximum icing conditions as described in Appendix C of Part 25 of this chapter.

(c) Compliance with all or portions of this section may be accomplished by reference, where applicable because of similarity of the designs, to analysis and tests performed for the type certification of a type certificated aircraft.

(d) When monitoring of the external surfaces of the airplane by the flight crew is required for proper operation of

the ice protection equipment, external lighting must be provided which is adequate to enable the monitoring to be done at night.

47. Paragraph (a) of § 23.1435 is amended to read as follows:

§ 23.1435 Hydraulic systems.

(a) *Design.* Each hydraulic system must be designed as follows:

(1) Each hydraulic system and its elements must withstand, without yielding, the structural loads expected in addition to hydraulic loads.

(2) A means to indicate the pressure in each hydraulic system which supplies two or more primary functions must be provided to the flight crew.

(3) There must be means to ensure that the pressure, including transient (surge) pressure, in any part of the system will not exceed the safe limit above design operating pressure and to prevent excessive pressure resulting from fluid volumetric changes in all lines which are likely to remain closed long enough for such changes to occur.

(4) The minimum design burst pressure must be 2.5 times the operating pressure.

48. Paragraph (c) (2) of § 23.1557 is amended to read as follows:

§ 23.1557 Miscellaneous markings and placards.

(c) *Fuel and oil filler openings.* . . .

(2) The word "oil" and the oil tank capacity and, for those aircraft for which no approved Airplane Flight Manual or approved manual material is required, the approved grade and specification of oil. (If an approved Airplane Flight Manual or approved manual material is provided, the approved grade and specification of oil may appear therein).

(Secs. 313(a), 601, and 603 of the Federal Aviation Act of 1958 (49 U.S.C. 1354, 1421, and 1423) and of section 6(c) of the Department of Transportation Act (49 U.S.C. 1655 (c).))

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JAMES E. DOW,
Acting Administrator.

November 28, 1973

Title 14—Aeronautics and Space
CHAPTER I—FEDERAL AVIATION ADMINISTRATION, DEPARTMENT OF TRANSPORTATION

[Docket No. 11011; Amdt. No. ~~22~~-14]

PART 23—AIRWORTHINESS STANDARDS:
NORMAL, UTILITY, AND ACROBATIC
CATEGORY AIRPLANES

Type Certification Standards

Correction

In FR Doc. 73-24430 appearing at page 31816, in the issue for Monday, November 19, 1973, the following changes should be made in § 23.155:

1. In paragraph (b)(2) the reference reading "V_{MO}/m_o" in the seventh line should read "V_{MO}/M_{MO}".

2. In paragraph (c) the words "stick force per gradient" in the fifth and sixth lines should read "stick force per g gradient".